

Microscale Phase Change Produces Inverse Cascade in Turbulent Clouds: Evidence from DNS

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Key Points:

- Turbulence-driven direct cascade and microphysics-driven inverse cascade intersect to cause a spectral break in the cloud water field.
- Across different turbulence intensities, the break occurs at a spatial wavelength of approximately eight times the Kolmogorov length.
- Continuous injection of microscale cloud water variance sustains spatial inhomogeneity and upscale transfer in turbulent clouds.

Abstract

The energy spectrum of cloud water content reveals a distinct break, occurring when turbulence-driven direct cascade is disrupted by another cascade. To understand the mechanism behind the second cascade, this study implements a particle-based direct numerical simulation approach to model entrainment-mixing in a turbulent cloud. All turbulence scales are explicitly resolved and cloud droplets are tracked in a Lagrangian fashion. It is shown that microscale evaporation of cloud droplets injects variance at the small scales, creating an inverse cascade in the cloud water field. The intersection of microphysics-driven inverse cascade and turbulence-driven direct cascade is marked by a spectral break whose physical-space length scales with the Kolmogorov length. The upscale transfer is fueled by the small-scale inhomogeneity of clouds, evidenced by the trimodal distribution of cloud water gradients. Moreover, the greater the difference between the spectral slopes across the break, the larger the broadening of cloud droplet size distributions.

1 Introduction

Atmospheric clouds regulate the Earth's radiative budget and hydrological cycle, yet their internal structure remains one of the difficult features to represent in weather and climate models. The key reason is the enormous range of scales involved. Cloud properties vary coherently from the kilometer scale of convection to the sub-millimeter scale at which individual droplets condense and evaporate. A recurring and informative signature of this multiscale behavior is the appearance of a break in the energy spectrum of cloud water field. This suggests that the power-law exponent in the Fourier space changes sharply and that the regimes on either side of the break are governed by different physical processes.

A closer look at the mechanisms behind the break reveals that microscale phase change influences liquid water content (LWC) spectrum. Davis et al. (1999) experimentally measured the LWC in clouds at approximately 4 cm resolution. The corresponding energy spectra showed two distinct power-law regimes, separated by a break. They proposed that an additional variance source exists at small scales but it remained unidentified because of insufficient measurement resolution. In order to theoretically analyze the spectral break, Jeffery (2001) solved an advection-diffusion equation for the LWC with inclusion of phase change. The author found that a production subrange is created in the LWC spectrum by condensation and evaporation. The suggestion that microscale phase change injects variance at small scales is supported by Gotoh et al. (2021). They theoretically showed that the LWC spectrum exhibits two scaling ranges as cloud growth or decay modifies the spectrum. Recently, Wang et al. (2026) observed a physical-space break in the size distribution of growing clouds. Based on large-eddy simulations (LES), they concluded that the break originated from dynamical cloud growth.

A break in the scaling laws can be caused by the intersection of direct and inverse energy cascades. In a laboratory investigation, Xia et al. (2011) showed that energy is transferred from the small to large scales in turbulence if a variance is externally imposed at the small scales. To distinguish between direct and inverse energy cascades, they calculated the third-order velocity structure function. Lilly (1983) suggested that Earth's mesoscale atmosphere has more energy than predicted by theory and the source of extra energy is upscale energy transfer. This hypothesis was tested by Vallis et al. (1997) who performed large-eddy simulations of mesoscale convection and confirmed the presence of inverse energy cascade, although the upscale transfer was reportedly smaller than the downscale transfer. Cabanes et al. (2020) reported that planetary rotation inhibits vertical motion and creates a stratified atmosphere in which energy is transferred from the small-scale sources up to the large-scale jets.

While the abovementioned studies identified upscale energy transfer and consequent break in the scaling behavior, they did not explore how turbulence-cloud microphysics interactions affect these phenomena. This study examines the transfer of energy across scales within a turbulent cloud. We apply a particle-based direct numerical simulation (DNS) model which resolves all scales of the flow and tracks cloud droplets individually in Lagrangian fashion. Our model setup is such that it simulates entrainment-mixing between the cloudy and clear air. The grid resolution is much finer than that used in the previous studies of this problem. The objective is to analyze how microscale phase change acts as a localized source of variance and transfers energy from the small to large scales. By contrasting simulations with and without cloud droplets, we isolate the contribution of phase change. We also vary the intensity of flow turbulence and assess its impact on the energy cascades.

2 Governing Equations

The turbulent flow of air is governed by the incompressible Navier-Stokes equations:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho_a} \nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{F}_b + \mathbf{F}_c, \quad (1)$$

$$\nabla \cdot \mathbf{u} = 0. \quad (2)$$

Above, $\mathbf{u}$ is the instantaneous air velocity vector, ν is the kinematic viscosity of air, p is the instantaneous air pressure, and ρ_a is the air density. $\mathbf{F}_c$ is an external force that maintains flow turbulence and is constructed in the Fourier space following the procedure of Eswaran and Pope (1988). The buoyancy force $\mathbf{F}_b$ is included but is too weak to sustain turbulence within a small domain. The conservation equations for the temperature (T) and water vapor mixing ratio (q_v) are:

$$\frac{\partial T}{\partial t} + (\mathbf{u} \cdot \nabla) T = \frac{L_h}{c_p} C_d + \mu_T \nabla^2 T + F_T, \quad (3)$$

$$\frac{\partial q_v}{\partial t} + (\mathbf{u} \cdot \nabla) q_v = -C_d + \mu_v \nabla^2 q_v + F_{q_v}. \quad (4)$$

Above, L_h is the specific latent heat, C_d is the time rate of exchange between water vapor and liquid water, and c_p is the specific heat of air. μ_T and μ_v are the molecular diffusivities while F_T , and F_{q_v} are the external forcing terms in the temperature and water vapor mixing ratio fields, respectively. The definitions of $\mathbf{F}_c$, $\mathbf{F}_b$, F_T , and F_{q_v} can be found in Sharfuddin et al. (2026). Note that F_T and F_{q_v} are necessary to sustain thermodynamic fluctuations in the respective fields. The environmental supersaturation (S_e) depends on the T and q_v , and is found as:

$$S_e = \frac{q_v}{q_{v,s}} - 1 = \frac{q_v}{621.97 \frac{p_{sat}}{p - p_{sat}}} - 1, \quad (5)$$

where $q_{v,s}$ is the saturation vapor mixing ratio and p_{sat} is the saturation vapor pressure which can be calculated with respect to water as: (Huang, 2018)

$$p_{sat} = 611.2 \times \exp\left(\frac{17.67(T - 273.15)}{T - 29.65}\right). \quad (6)$$

The equations for the Lagrangian position ($\mathbf{X}$) and velocity ($\mathbf{V}$) vectors of cloud droplets are written as:

$$\frac{d\mathbf{X}_i(t)}{dt} = \mathbf{V}_i(t), \quad \frac{d\mathbf{V}_i(t)}{dt} = \frac{1}{(\tau_p)_i} [\mathbf{u}_i(\mathbf{X}, t) - \mathbf{V}_i(t)] + \mathbf{g}, \quad (\tau_p)_i = \frac{2\rho_l(r_i)^2}{9\rho_a\nu}. \quad (7)$$

where the subscript ‘ i ’ denotes the i -th particle, r is the particle radius, $\mathbf{u}_i$ is the instantaneous air velocity vector interpolated at the position of particle i , $\mathbf{g}$ is the gravity vector, τ_p is the particle response time, and ρ_l is the density of liquid water. Cloud droplets grow or shrink according to the following equation:

$$\frac{d(r(t))_i}{dt} = \left(\frac{G}{r(t)} S_e(\mathbf{X}, t) \right)_i, \quad G = \frac{1}{\frac{L_h \rho_l}{k_T T} \left(\frac{L_h}{R_v T} - 1 \right) + \frac{\rho_l R_v T}{\mu_v p_{sat}}}. \quad (8)$$

where G is the growth factor, k_T is the thermal conductivity in air, and R_v is the universal gas constant divided by the molecular weight of water. Then, C_d is determined as:

$$C_d(\mathbf{x}, t) = \frac{4\pi \rho_l G}{\rho_a a^3} \sum_{i=1}^n (r(t) S_e(\mathbf{X}, t))_i, \quad (9)$$

where n is the number of cloud droplets within a grid cell having a volume of a^3 . The curvature and solute effects are neglected in the present study because cloud droplets remain much larger than dry aerosols throughout the simulation period. Finally, the liquid water mixing ratio q_l is found as:

$$q_l(\mathbf{x}, t) = \frac{4\pi \rho_l}{3\rho_a a^3} \sum_{i=1}^n r_i^3(t). \quad (10)$$

The q_l denotes cloud (liquid) water content and is directly affected by condensational growth or evaporative decay of cloud droplets.

3 Numerical Details

Flow turbulence is assumed to be homogeneous and isotropic. We generate three turbulent kinetic energy (TKE) dissipation rates by varying the band of forcing wavenumbers. We denote the three TKE levels as ‘low’ (L), ‘medium’ (M), and ‘high’ (H), respectively. The corresponding wavenumber bands are: $1 \leq k_f/k_{min} \leq 2.45$, $1 \leq k_f/k_{min} \leq 4.12$, and $1 \leq k_f/k_{min} \leq 13.86$ while the Kolmogorov length scales (η) are: 0.0019 m, 0.0015 m, and 0.00088 m. Note that $k_{min} = 2\pi/L = 12.27$ [1/m] is the minimum wavenumber and k_f is the forcing wavenumber. The external thermodynamic forcing terms (F_T and F_{q_v}) are constructed in the Fourier space with a wavenumber band of $1 \leq k_f/k_{min} \leq 4.12$. More details about external forcing and our particle-based DNS model can be found in Sharfuddin et al. (2026).

The physical model is a cube and has a volume of 0.512^3 m³, with each side having $L = 0.512$ m. The boundary conditions are triply periodic. The domain is equally divided into cloudy and clear-air regions at the start. A slab-like configuration is assumed so that the initial temperature and water vapor mixing ratio change sharply at the cloud/clear-air interface (Gao et al., 2018). The initial vapor mixing ratio is specified as:

$$q_v(z, t = 0) = \begin{cases} q_v^{max}, & d - L \times 0.5 \times 0.5 \leq z < d + L \times 0.5 \times 0.5 \\ q_{v,e}, & \text{elsewhere} \end{cases}. \quad (11)$$

Above, $q_v^{max} = 3.949$ g/kg, $q_{v,e} = 0.3$ g/kg, and $d \equiv L/2$ is the width of the cloud slab. q_v^{max} and $q_{v,e}$ represent cloudy (supersaturated) and dry (subsaturated) regions, respectively. The initial temperature field is prescribed as:

$$T(t = 0) = \langle T(t = 0) \rangle - 0.608 \langle T(t = 0) \rangle (q_v(z, t = 0) - \langle q_v(t = 0) \rangle). \quad (12)$$

We simulate four cases: H, H0, M, and L. The details are provided in Table 1 along with the values of simulation parameters. 16×10^6 cloud droplets are initially placed in the cloudy region, giving a droplet number density of 119.2 cm^{-3} . Entrainment of dry air and turbulent mixing spread cloud droplets to all over the domain. The dominant phase change process is evaporation. The number of grid points is 256^3 and the grid resolution is 2 mm for all cases. We run simulations for a physical time of 20 s. The $k_{max}\eta$ value is 1.38 for Cases H and H0, 2.36 for Case M, and 2.98 for Case L, where $k_{max} = (256\pi)/L = 1570.27 \text{ [1/m]}$ is the maximum wavenumber. The $k_{max}\eta$ measures how accurately the viscous dissipation range is captured in a simulation and is prescribed to be at least greater than 1.0 (Pope, 2000).

Table 1*Details of the test cases*

Case	TKE dissipation rate ϵ_{TKE} (m^2/s^3)	Level of momentum forcing $\mathbf{F}_c$	Initial radius of each cloud droplet (μm)	Stokes number $S_t = \tau_p / \sqrt{(v/\epsilon_{TKE})}$ at initial time	Inclusion of droplets
H	0.0056	High	15	$6.42e - 2$	Yes
M	0.0006	Medium	15	$2.12e - 2$	Yes
L	0.00025	Low	15	$1.36e - 2$	Yes
H0	0.0056	High	15	N/A	No

Case	Description
H	high (H) level of flow turbulence, cloud droplets included
M	medium (M) level of flow turbulence, cloud droplets included
L	low (L) level of flow turbulence, cloud droplets included
H0	high (H) level of flow turbulence, cloud droplets excluded (0)

Parameters and their values

Symbol	Value	Unit	Symbol	Value	Unit
L_h	2.5×10^6	J/kg	$\mu_T = \mu_v$	2.2×10^{-5}	m^2/s
ρ_l	1000	kg/m^3	c_p	1005.0	J/kg/K
$p(t = 0)$	71446	N/m^2	$\langle T(t = 0) \rangle$	270.75	K
ρ_a	0.92	kg/m^3	ν	1.5×10^{-5}	m^2/s
R_v	461.5	J/kg/K	k_T	0.0238	W/m/K

4 Results and Discussion

This section presents the findings from our particle-based direct numerical simulations. We focus on a single timeframe at 10 s, when velocity and thermodynamic fluctuations are statistically stationary. Because the primary insights of this study are spatial in nature, analyzing one temporal

snapshot proves sufficient for our analysis. We investigate the energy spectra, scale-dependent skewness and spatial derivatives of key quantities, as well as cloud droplet size distributions.

The energy spectra are shown in Figures 1(a)-(f). The E_{TKE} is higher at the large scales for Cases H and H0. This is expected because only the low-wavenumber region is forced. The four E_{TKE} curves converge at the intermediate scales. Difference is visible in the dissipation range between high (H and H0) and medium/low (M/L) turbulence cases because $k_{max}\eta$ is smaller for the former. The TKE spectra (Figure 1(a)) are identical for Cases H and H0, suggesting that cloud droplets do not influence flow turbulence. There appears to be a reduction of downscale transfer in the temperature and water vapor spectra at the small scales (Figures 1(b)-(c)) if cloud droplets are present (Case H, M, and L). Whereas in the absence of microphysical feedback (Case H0), E_T and E_{qv} decay smoothly. To show the impact of phase change on the energy spectra, we analyze E_{Cd} , E_{ql} , and E_r in Figures 1(d)-(f), respectively. It is evident that microscale evaporation causes upscale energy transfer in the C_d (Figure 1(d)) and the slope of E_{Cd} turns positive before the final decay. The inverse (upscale) cascade intersects with the direct (downscale) cascade. A break is observed in the scaling behavior of E_{Cd} because the slope is negative (-2.85) to the left of the intersection but positive (1.2) to the right. A negative slope is indicative of turbulence-dominated direct cascade while a positive slope implies a microphysics-dominated inverse cascade. The E_{ql} peaks in the high-wavenumber region as phase change injects energy at the small scales. If k_b is the wavenumber corresponding to the break, the physical-space break occurs at wavelength $\lambda_b = (2\pi/k_b)$. Based on Figure 1(e), $\lambda_b = (2\pi/(10^{-0.2}/0.00088)) \sim 8.76 \text{ mm}$, $\lambda_b = (2\pi/(10^{-0.15}/0.00153)) \sim 13.48 \text{ mm}$, and $\lambda_b = (2\pi/(10^{-0.05}/0.0019)) \sim 13.39 \text{ mm}$ for Cases H, M, and L, respectively. Interestingly, the value of λ_b/η is ~ 9.95 for Case H, ~ 8.8 for Case M, and ~ 7.05 for Case L. Hence, a break is located in the cloud water field at $\lambda_b \sim 8\eta$ between the inertial and dissipation regions (Jeffery, 2001). The break is directly related to the Kolmogorov scale, showing its dependency on the intensity of flow turbulence. We see that the right slope is positive (2.6) for all three cases (Figure 1(e)) and corresponds to the high wavenumber region. The left slope, which corresponds to the intermediate wavenumber region, is found to be negative for Cases L (-0.6) and M (-0.5) but positive (0.45) for Case H. A direct cascade (negative slope) exists in the cloud water field for low and medium turbulence levels but is overwhelmed by the inverse cascade (positive slope) when the turbulence level is high. The direct cascade is weaker and the E_T , E_{qv} , E_{Cd} , and E_{ql} contain less energy for Case H, as indicated by a smaller area under the curve in Figures 1(b)-(e). A high flow turbulence level rapidly smooths out thermodynamic and microphysical gradients and reduces the spatial variance (energy). The radius spectra (E_r) in Figure 1(f) reaffirm the energy injection at high wavenumbers and upscale transfer. Also, a direct cascade is present in the low wavenumber region of E_r , as shown by the local peaks at $k\eta \sim 10^{-1.7}$, $k\eta \sim 10^{-1.5}$, and $k\eta \sim 10^{-1.4}$ for Cases H, M, and L, respectively. Since the supersaturation and particle fields are coupled (Equation (8)), large-scale fluctuations in the S_e are transmitted to the particle radius.

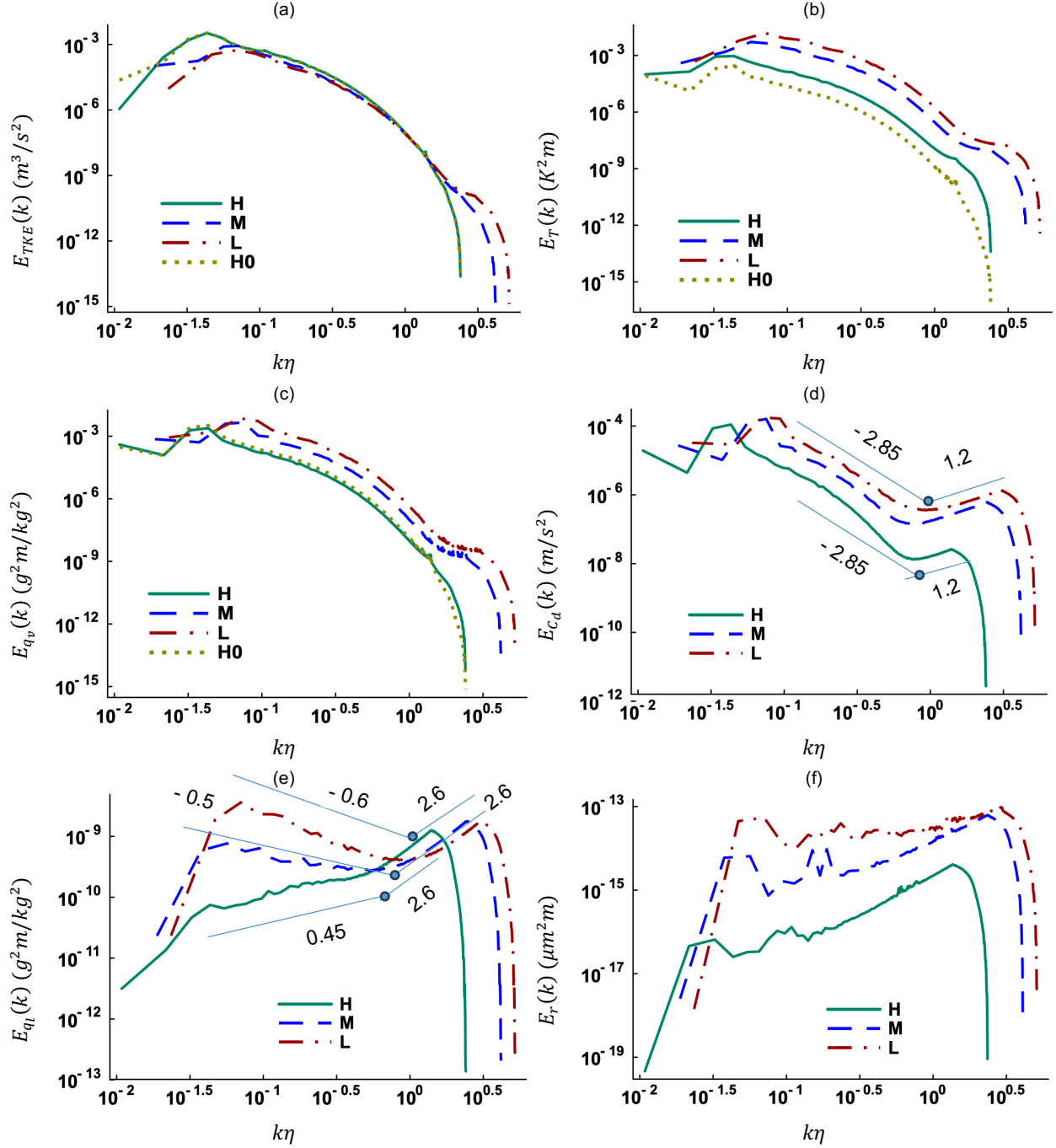


Figure 1. Energy spectra of the (a) turbulent kinetic energy (E_{TKE}), (b) temperature (E_T), (c) water vapor mixing ratio (E_{qv}), (d) phase change term (E_{Cd}), (e) liquid water mixing ratio (E_{ql}), and (f) particle radius (E_r) at 10 s. The intersection points in Figures 1(d)-(e) show spectral breaks.

The scale-dependent normalized skewness is calculated as (Garg et al., 2021)

$$\gamma_\phi(\Delta z) = \frac{\langle (\phi(z + \Delta z) - \phi(z))^3 \rangle}{\langle (\phi(z + \Delta z) - \phi(z))^2 \rangle^{3/2}}. \quad (13)$$

This normalized third-order structure function measures the asymmetry of increments and hence the direction of energy transfer. A zero value indicates no net transfer, while a negative sign implies a direct cascade and a positive sign suggests an inverse cascade (Frisch, 1995). The results for vertical velocity w and cloud water mixing ratio q_l are shown in Figures 2(a)-(b), respectively. We observe that γ_w is highly negative at small separations and approaches zero as separation distance increases (Figure 2(a)). Therefore, energy flows steadily from the large to small scales, with no indication of upscale transfer in the momentum field. Ideally, γ_w should vanish at large separations as widely separated points become statistically independent. In our simulations, the domain length (L) is 0.512 m and the integral length scale l is approximated as 0.1 m, 0.15 m, and 0.2 m for Cases H, M, and L, respectively. Because L/l is smaller than the theoretically recommended value of 8 (Pope, 2000), velocity increments at the large scales do not reach full statistical independence and a small residual skewness exists at large Δz . This residual does not affect the small-scale behavior, which is of interest in our study.

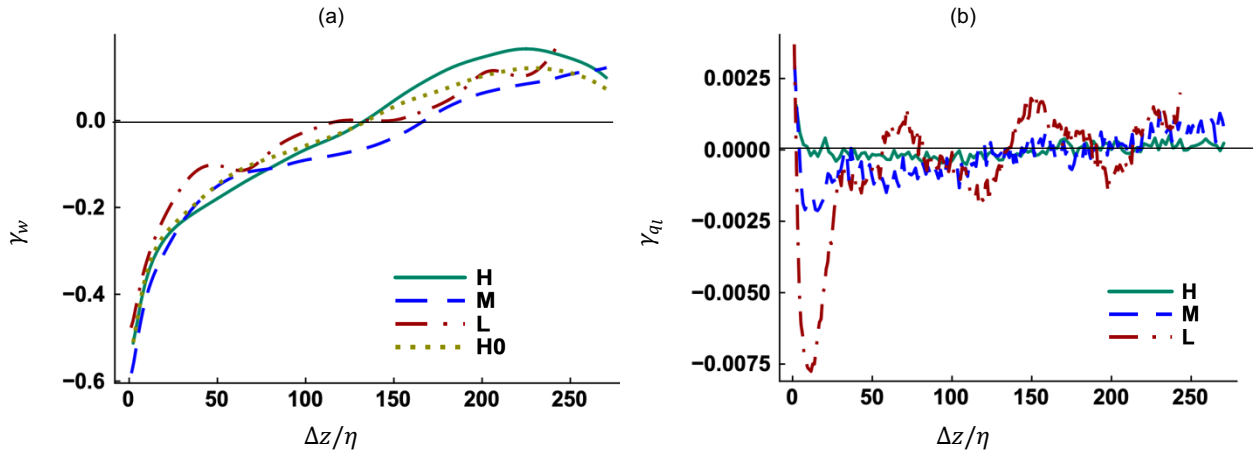


Figure 2. Scale-dependent skewness for the (a) z -velocity (γ_w) and (b) cloud water mixing ratio (γ_{q_l}) at 10 s.

The increments in cloud water content behave differently. Figure 2(b) shows that γ_{q_l} is slightly positive at a very small Δz for Cases H, M, and L. The positive skewness is consistent with the inverse cascade observed in the E_{q_l} (Figure 1(e)). As Δz increases, the direct cascade starts to dominate and γ_{q_l} becomes negative for Cases M and L before approaching zero at large separations. However, γ_{q_l} does not become meaningfully negative for Case H at any separation. This is again consistent with Figure 1(e) which shows that direct cascade is present for low and medium turbulence levels but not for the high turbulence level. Hence, the third-order statistics in physical space confirm that turbulence-dominated direct cascade meets microphysics-dominated inverse cascade in the cloud water field. The separation at which γ_{q_l} crosses the zero line (Figure 2(b)) is calculated as $\sim 1.62\eta = 3.1$ mm for Case L, $\sim 4.09\eta = 6.1$ mm for Case M, and $\sim 9.78\eta = 8.6$ mm for Case H. So, the crossover length is larger in response to high flow turbulence and reduced cloud water variance (Case H). It appears that the γ_{q_l} profiles are noisy while the γ_w curves are smooth. The noise stems from the discrete, particle-based representation of cloud droplets.

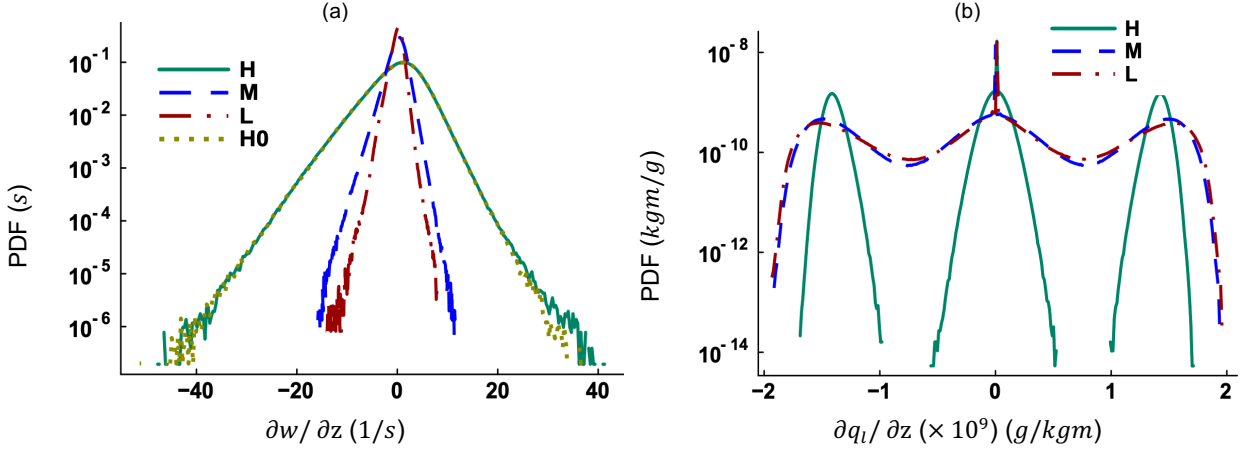


Figure 3. Probability density function (PDF) of the (a) vertical velocity derivative and (b) cloud water mixing ratio derivative at 10 s. The derivatives are calculated in the z -direction using the central difference formula. Note that the PDFs are identical if the derivatives are calculated in the x - or y - coordinate direction.

Spatial derivatives quantify the local steepness of a turbulent field. The vertical velocity derivative $\partial w/\partial z$ in Figure 3(a) suggests that strong turbulence (H and H0) extends the tail of the distribution because of more energy and intermittency. It also shows that the distribution is nearly symmetric, with equal magnitude of positive and negative gradients. The distribution of $\partial q_l/\partial z$ in Figure 3(b) is trimodal for all three cases (H, M, and L) permitting phase change. The domain begins with equal volumes of cloudy and clear air, and subsequent mixing produces interleaved pockets of cloudy (supersaturated) and clear (subsaturated) air. The central peak corresponds to the interior of either pocket, where $\partial q_l/\partial z$ is essentially zero because q_l is nearly uniform in the cloudy pocket and zero in the clear air pocket. The left peak (negative $\partial q_l/\partial z$) marks the transition from a cloudy pocket outward into clear air, where the liquid water content falls sharply along the z direction. The right peak (positive $\partial q_l/\partial z$) marks the reverse transition from clear air into cloud. The $\partial q_l/\partial z$ varies gradually for Cases L and M, and in discrete jumps for Case H. Thus, the side peaks become more pronounced as turbulence intensifies. The sharp changes in q_l suggests the persistence of small-scale structures which inject variance at high wavenumbers. This reflects a basic difference between the velocity and cloud water fields. Within the velocity field, molecular diffusion continuously smooths out spatial gradients, ensuring that their derivatives maintain finite, well-defined variances balanced between turbulent production and diffusive decay. In contrast, the cloud water mixing ratio lacks an analogous smoothing mechanism. Because it is constructed using discrete droplet positions (Equation (10)), it is not subject to molecular diffusion. This absence of diffusive smoothing allows the cloud water to sustain variance at small scales (high wavenumbers), ultimately fueling the inverse cascade responsible for the spectral break observed in Figure 1(e).

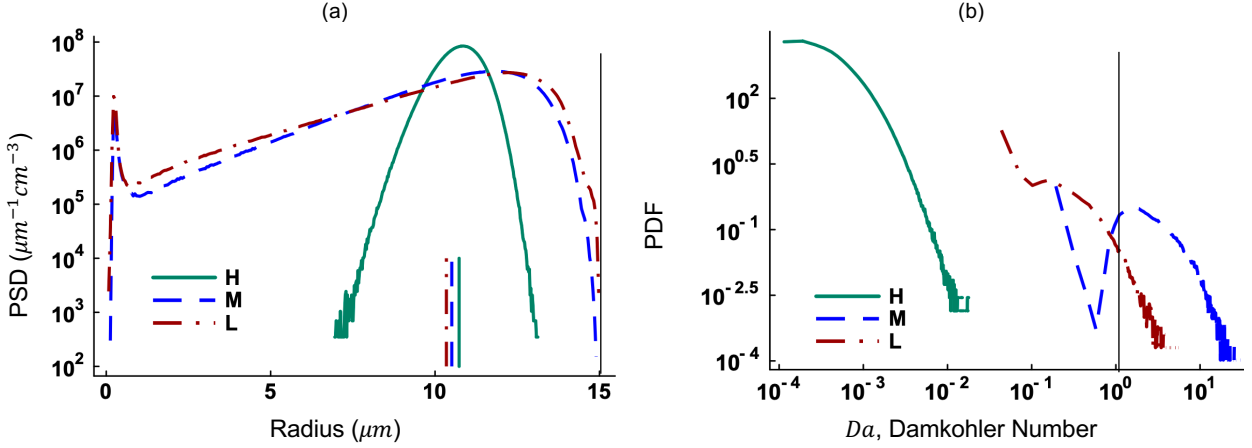


Figure 4. (a) Particle size distribution (PSD) of cloud droplets and (b) probability density function (PDF) of Damkohler number at 10 s. Note that $\text{PSD} = \text{PDF} \times \text{number concentration}$. The vertical line segments in subplot (a) show the expectation (mean) of cloud droplet radii. The solid vertical line in subplot (a) shows the initial monodisperse ($15 \mu\text{m}$) size distribution. Also, the unity Damkohler number is marked by a solid vertical line in subplot (b).

Flow turbulence impacts the size distribution of cloud droplets, as shown in Figure 4(a), even though the mean radii are very close. Beginning from a monodisperse population of $15 \mu\text{m}$ droplets, entrainment and evaporation broaden the distribution and shift its mode toward smaller radii as cloud droplets lose mass in subsaturated pockets. The broadening is stronger for Cases L and M and weaker for Case H. Higher turbulence homogenizes the supersaturation field rapidly and causes droplets to evaporate uniformly. Hence, the PSD is relatively narrower for Case H. On the other hand, weaker and localized mixing in Cases L and M exposes cloud droplets to uneven supersaturation levels and broadens the radius distribution. The spatial dispersion of cloud droplet radii is the origin of variance in q_l , which is the cubic moment of droplet radius (Equation (10)) and inherits the small-scale inhomogeneity of the droplet population. The microphysical dispersion cannot be accounted for in the spatially averaged, coarse-resolution measurements of Davis et al. (1999), necessitating our particle-based DNS approach. We find that a greater difference between the slopes on the opposite sides of the break (Figures 1(e)) correspond to a larger broadening of the PSD (Figures 4(a)). Since measuring the bulk cloud water content is easier with instruments like hot-wire probes and resolving the PSD requires expensive optical spectrometers, the spectral slopes of the easily measurable q_l might serve as a proxy for inferring the broadening of cloud droplet size distributions.

The PSD, as evaluated in a Lagrangian framework, appears to be bimodal when turbulent mixing is weaker (Cases L and M). To understand the mixing regime, we calculate the Damkohler number Da as (Lehmann et al., 2009)

$$Da(\mathbf{x}, t) = \frac{\tau_\eta(\mathbf{x}, t)}{\tau_m(\mathbf{x}, t)}, \quad \tau_{\text{evap}}(\mathbf{x}, t) = \frac{(r(t))^2}{|GS_e(\mathbf{x}, t)|}. \quad (14)$$

$$\tau_\eta(\mathbf{x}, t) = \sqrt{\frac{\nu}{\epsilon_{TKE}(\mathbf{x}, t)}}, \quad \epsilon_{TKE}(\mathbf{x}, t) = \nu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right). \quad (15)$$

where ϵ_{TKE} is the TKE dissipation rate, τ_η is the Kolmogorov timescale (Pope, 2000), and τ_{evap} is the evaporation timescale. Figure 4(b) shows that Da can be larger than unity for Cases L and

M but remains below unity for Case H. $Da > 1$ suggests that evaporation outpaces turbulent mixing and cloud droplets grow or shrink based on local supersaturation before being carried away to another supersaturation pocket. This creates bimodality (Manton, 1979) as some cloud droplets completely evaporate while other droplets hardly evaporate (Figure 4(a)). Conversely, $Da < 1$ implies that turbulent mixing outpaces evaporation. Such rapid mixing forces all cloud droplets to evaporate at nearly identical rates and maintains a unimodal distribution. Whether the Damkohler number exceeds or falls below unity reflects the modality of cloud droplet size distribution in our problem.

5 Implications and Concluding Remarks

The DNS results presented in Section 4 carry several implications for the modeling and observation of cloud microphysics, which are summarized below.

- A) The upscale transfer in the cloud water mixing ratio (Figures 1(e)-(f)) originates below the resolution of mesoscale and global models. So, these models must account for the injection of small-scale variance in the cloud water field through improved parameterization.
- B) The scale-dependent skewness of q_l is positive at very small spatial separations and crosses the zero line at ~ 3 mm and ~ 9 mm for low and high turbulence levels, respectively (Figure 2(b)). The grid spacing in a numerical model should be such that this crossover is captured.
- C) The trimodal distribution of $\partial q_l / \partial z$ (Figure 3(b)) suggests that cloudy and clear-air pockets exist within a cloud. Even though turbulent mixing is active, microscale phase change injects variance continuously and maintains sharp gradients. Therefore, the spatial inhomogeneity in atmospheric clouds must be accurately represented.
- D) The broadening of cloud droplet size distribution can be inferred from the Eulerian evaluation of q_l . But a Lagrangian description is needed to capture the modality of radius distributions (Figure 4(a)).

We conclude that microscale phase change acts as a localized energy source within clouds and transfers cloud water variance upward. Capturing this phenomenon requires a particle-based DNS model which calculates phase change at precise droplet locations and prevents numerical diffusion from artificially destroying local gradients that fuel the inverse cascade. However, these conclusions rest on the idealized assumption of homogeneous and isotropic turbulence while turbulence in real clouds is inhomogeneous and anisotropic due to buoyancy and shear. Future iterations of this particle-based DNS model will implement non-periodic boundary conditions to account for buoyancy and shear effects and investigate the energy cascades in real clouds.

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